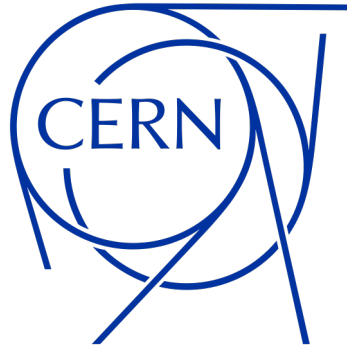
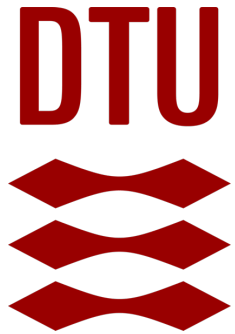




# Internship project Status W15: CAD Import & meshing in Wakis

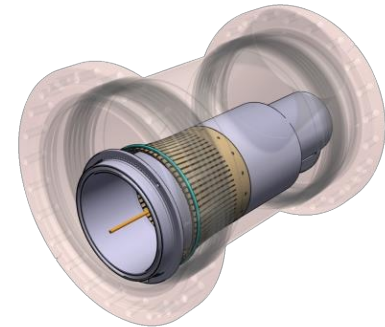
**Student:** Clara R.Wimmelmann. **Supervisors:** Elena De La Fuente Garcia, Carlo Zannini & Jakob Schiøtz

**Date:** 13/5-2026

# Outline

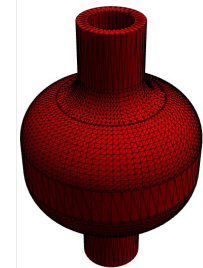
## 1. HTCondor simulation of the 6L2

- Bellows & fingers meshing issues
  - mask slices (qualitative: visual inspection)
  - Wakis simulation results
- **Closed transition tube** of the 6L2 meshing and simulation results



## 2. Perfect Boundary Approximation (PBA) / sub-cell smoothing benchmarks on shell

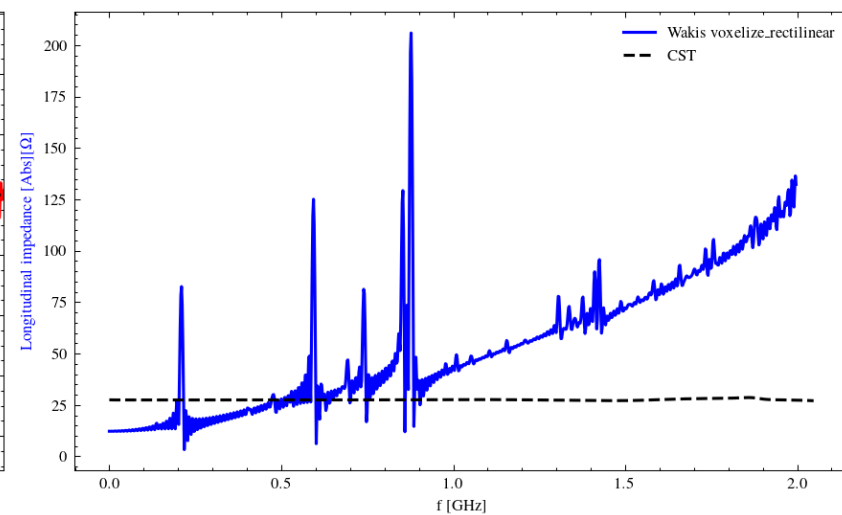
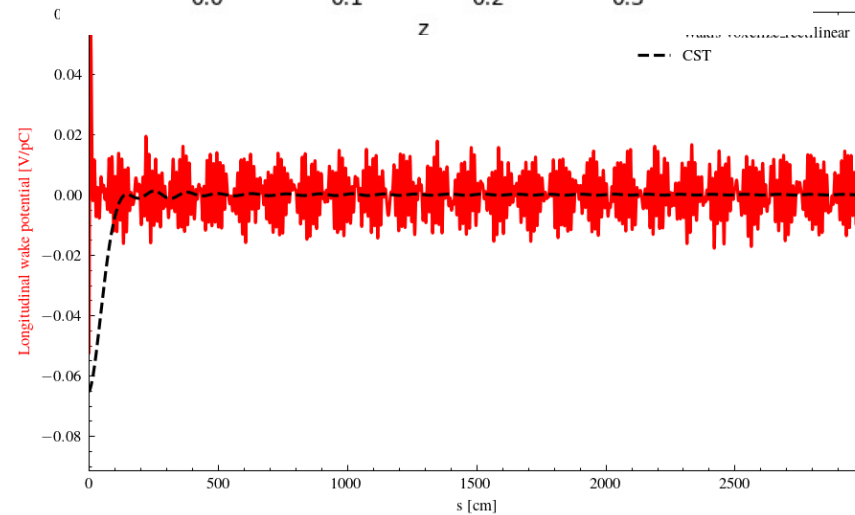
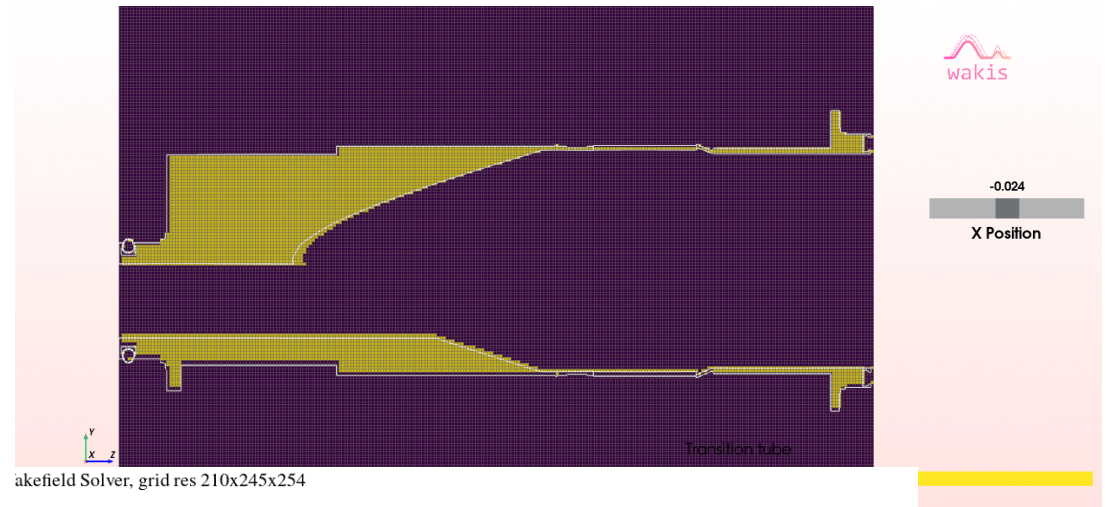
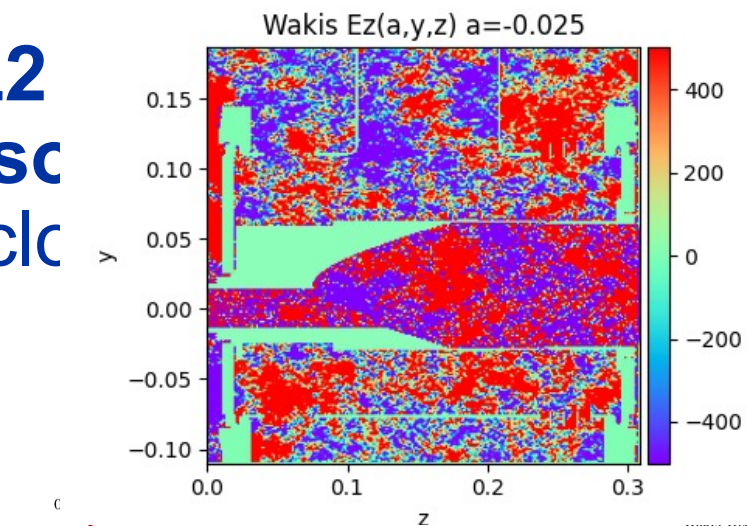
- PyVista .clip\_surface
  - Attributes of the function; value, +/- implicit distance
  - 6L2
  - Shell benchmarks



## 3. Aim of this week:

- Get stable Wakis simulation of the shell-cavity with PBA mask
  - Then: allows us to move forward with 6L2

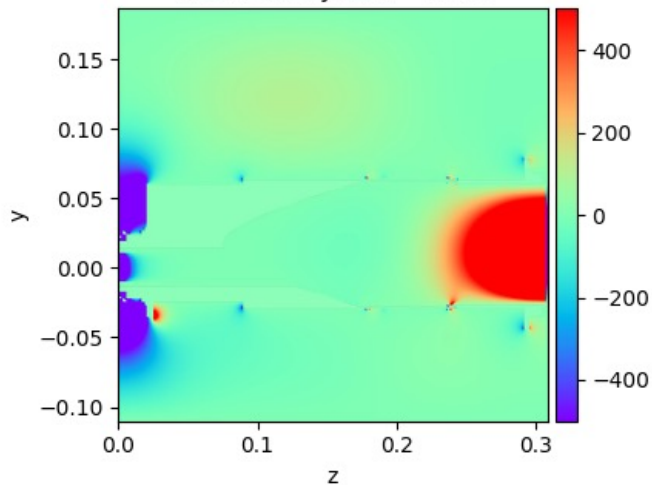
# 6L2 resc a clc



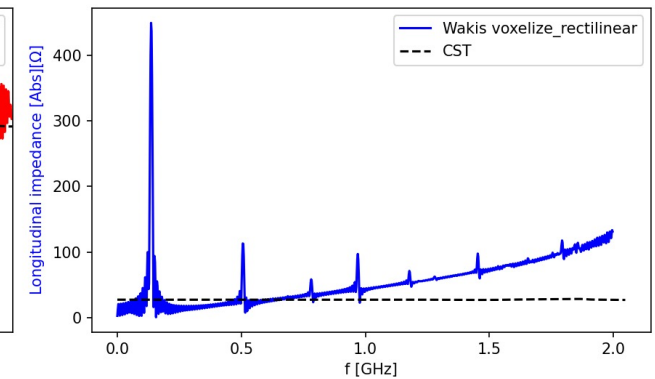
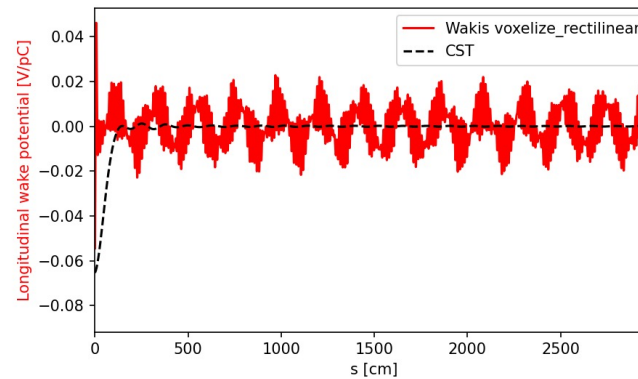
# closed Transition tube (in meters) **medium grid resolution benchmark: still leaks**

$E_z$  field, timestep=2100

Wakis  $E_z(a,y,z)$   $a=-0.025$

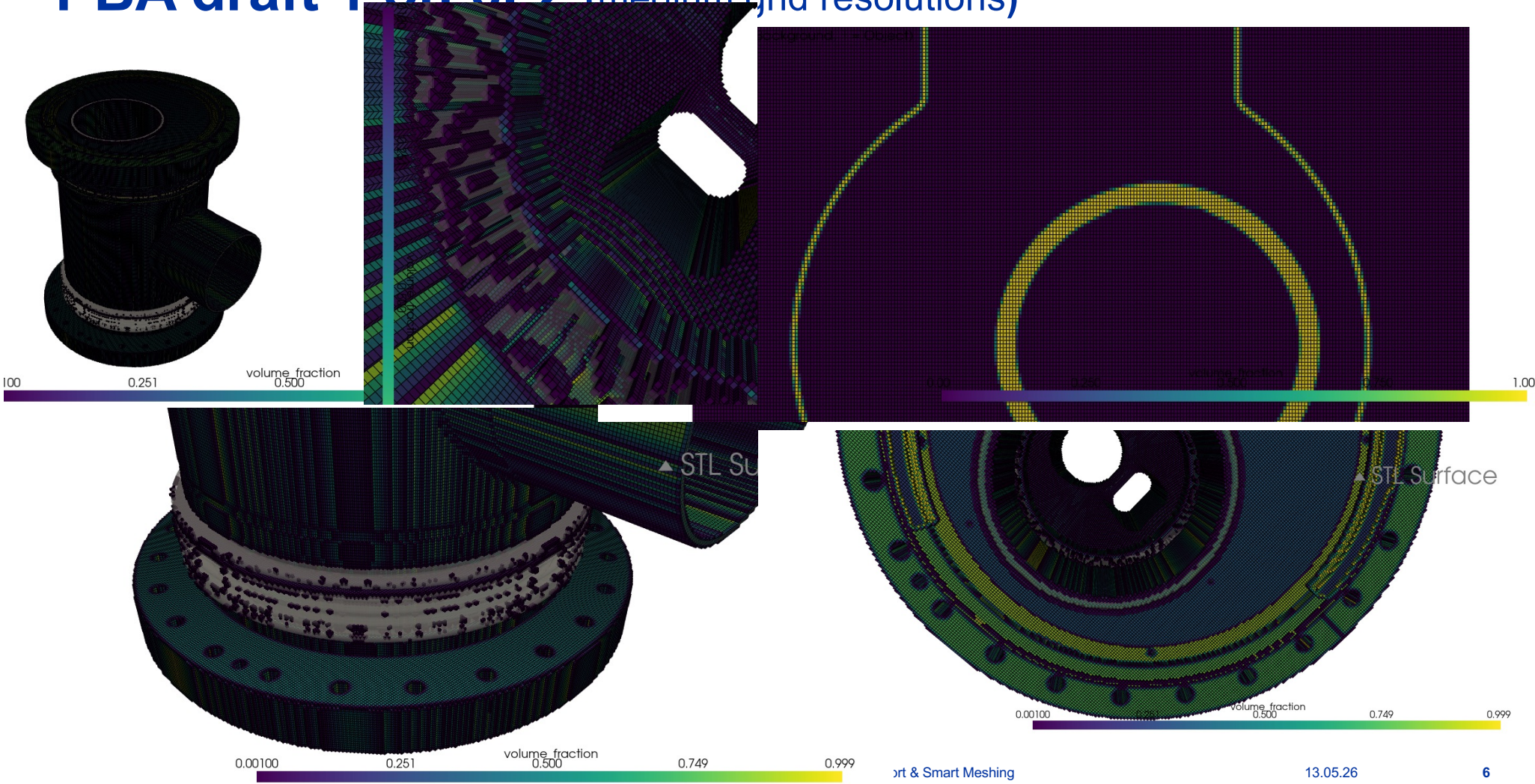


Benchmark with CST Wakefield Solver, grid res 210x245x254





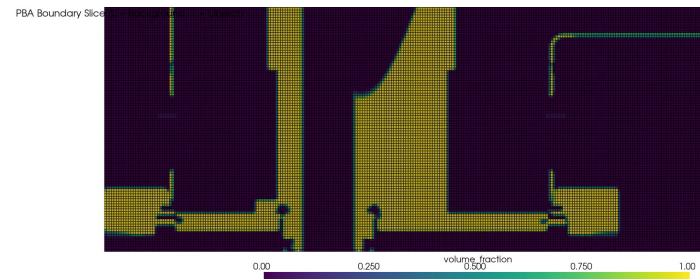
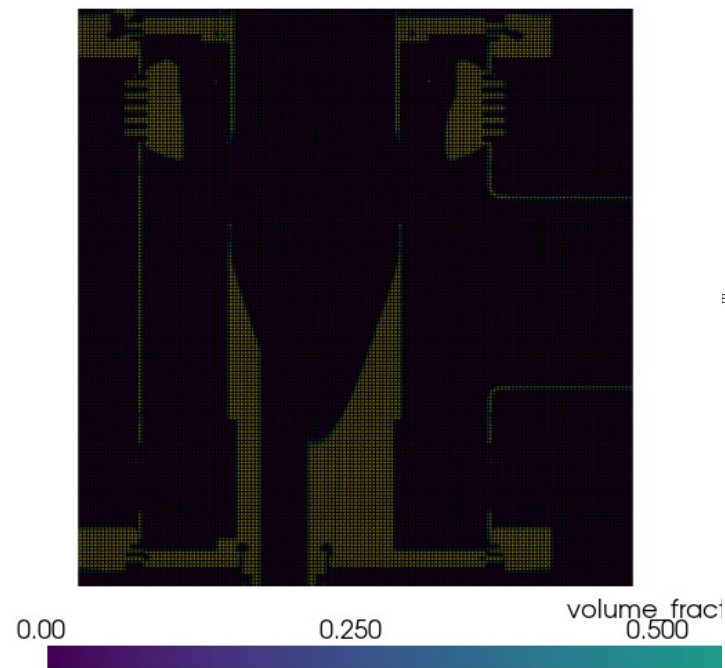
# PBA draft 1 on 6l 2 (medium grid resolutions)



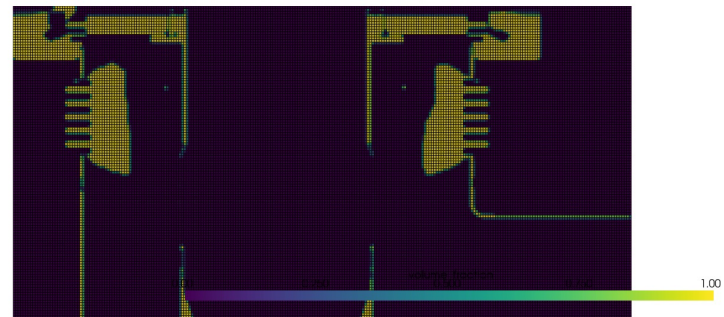


# PBA draft 1 on 6L2 (medium grid resolutions)

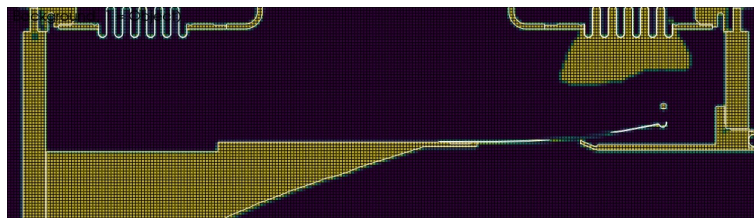
= Object)



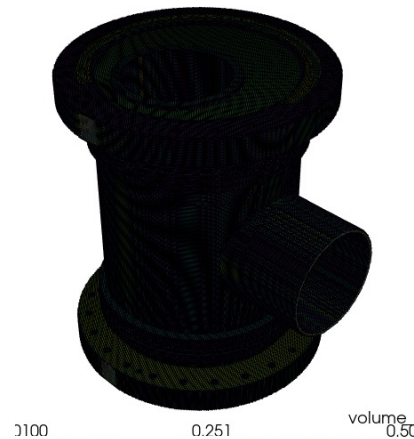
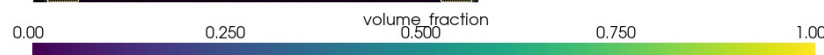
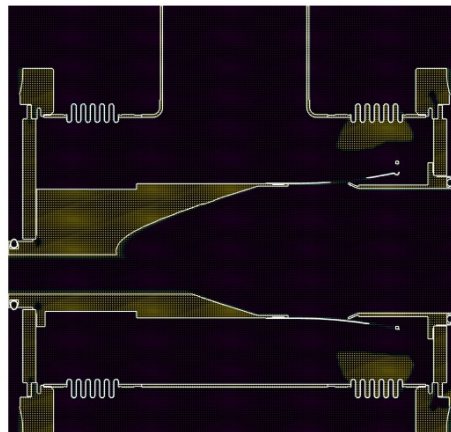
BA Boundary Slice (0 = Background, 1 = Object)



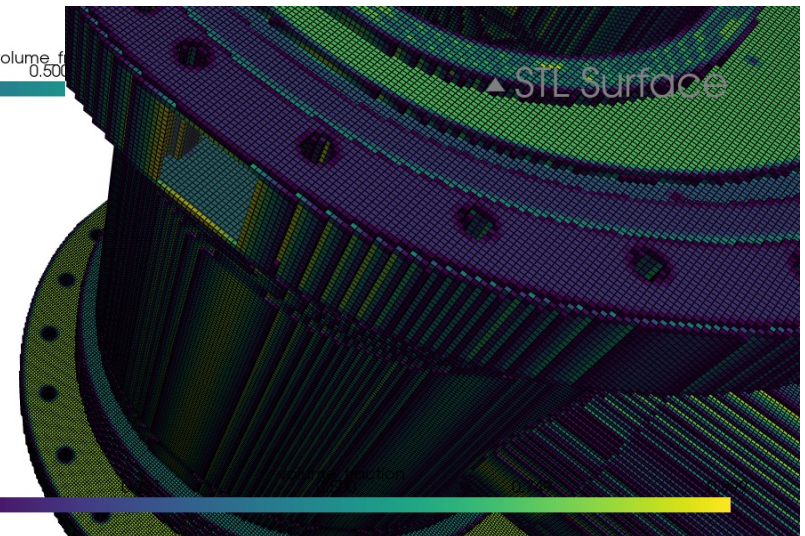
# PBA draft 2 on 6L2 (medium grid resolutions): using distance measure and threshold value



= Background, 1 = Object)



0.00 0.251 0.500  
volume fraction

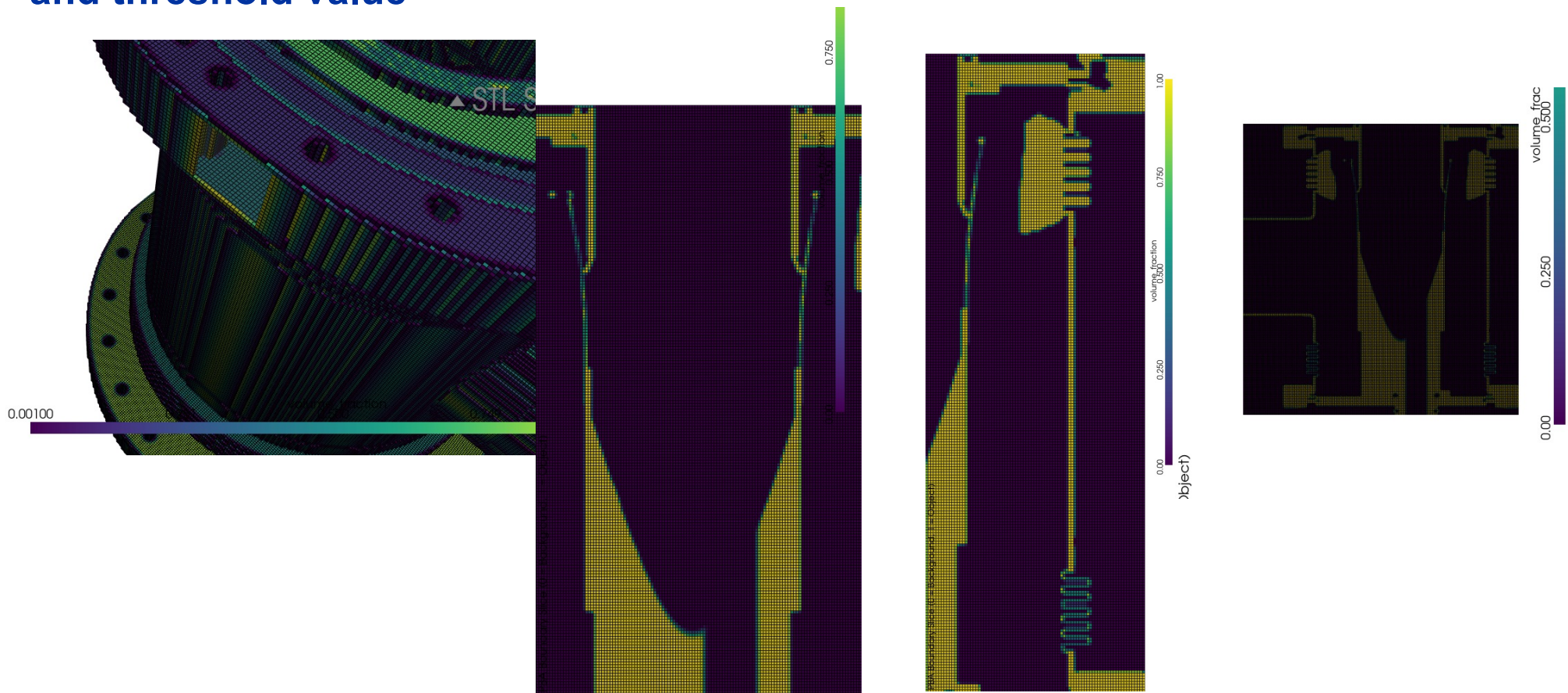


0.00100 0.500 1.00  
volume fraction

▲ STL Surface



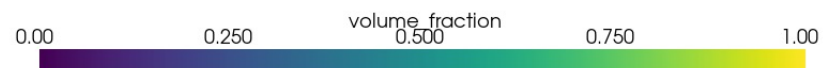
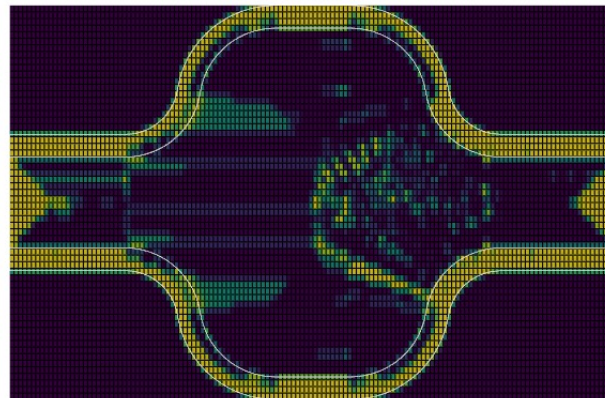
# PBA draft 2 on 6L2 (medium grid resolutions): using distance measure and threshold value





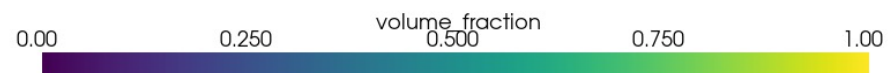
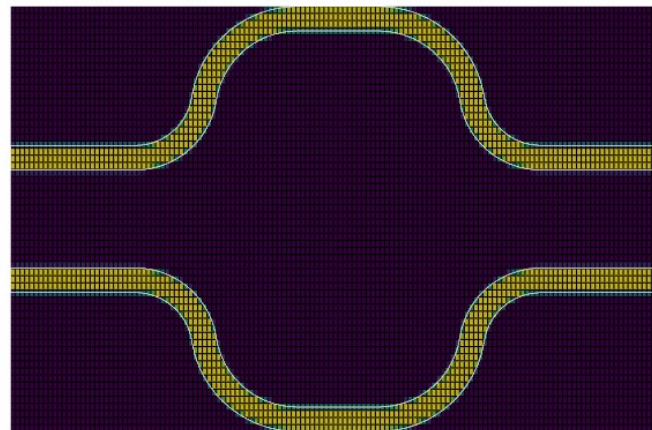
# PBA shell 1 : qualitative/mask

PBA Boundary Slice (0 = Background, 1 = Object)

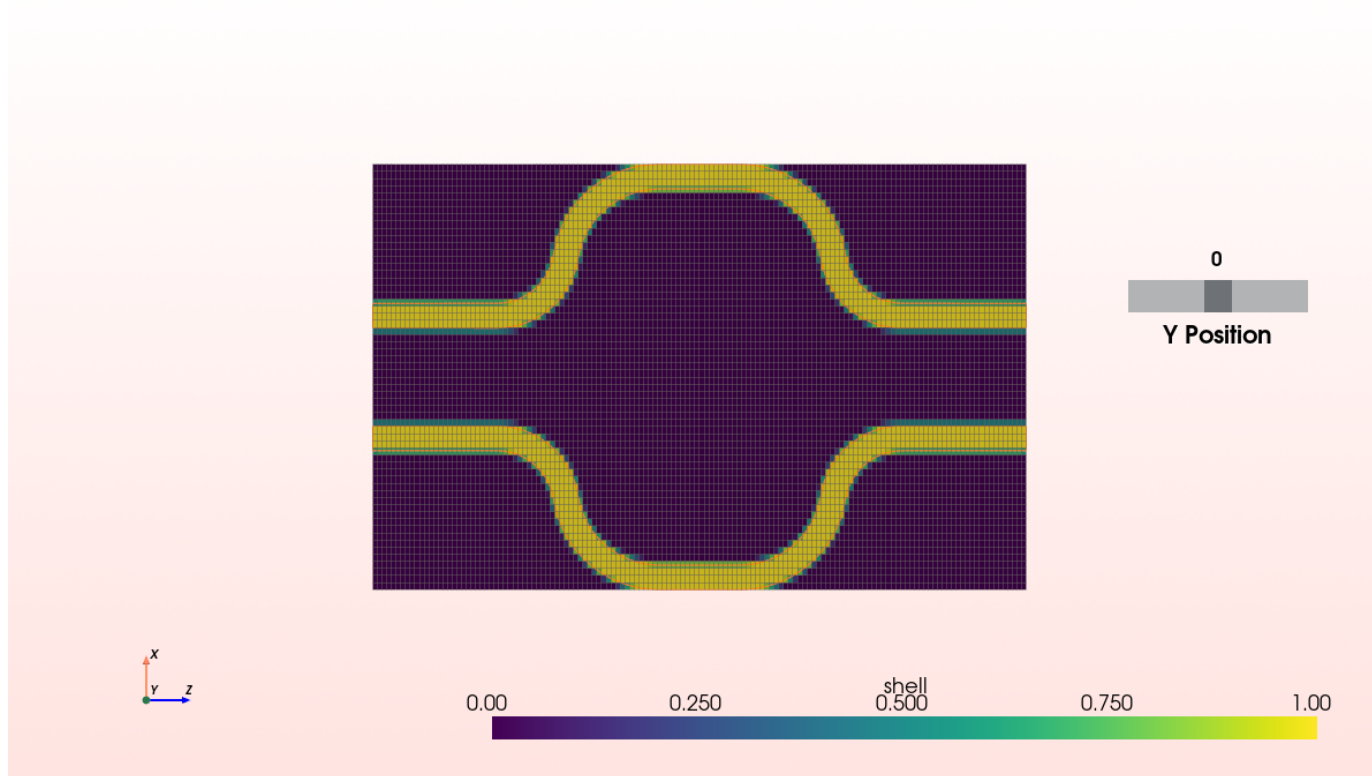


# PBA shell 2 : qualitative/mask

PBA Boundary Slice (0 = Background, 1 = Object)



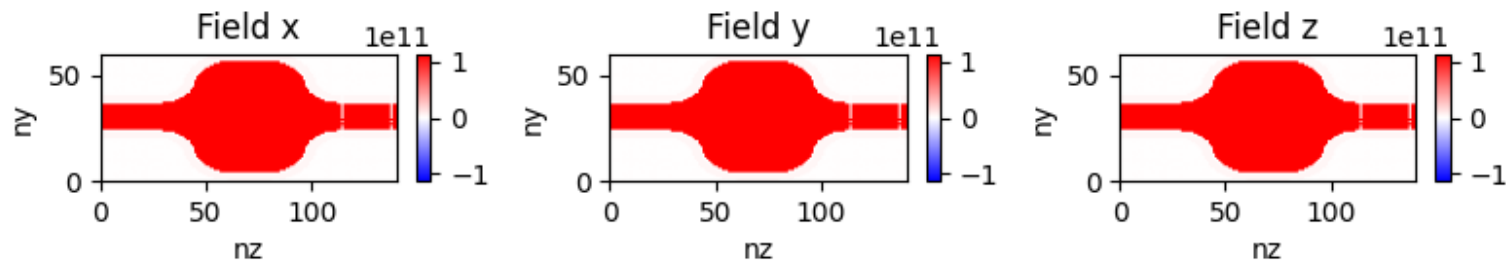
## PBA shell 3 : qualitative/mask



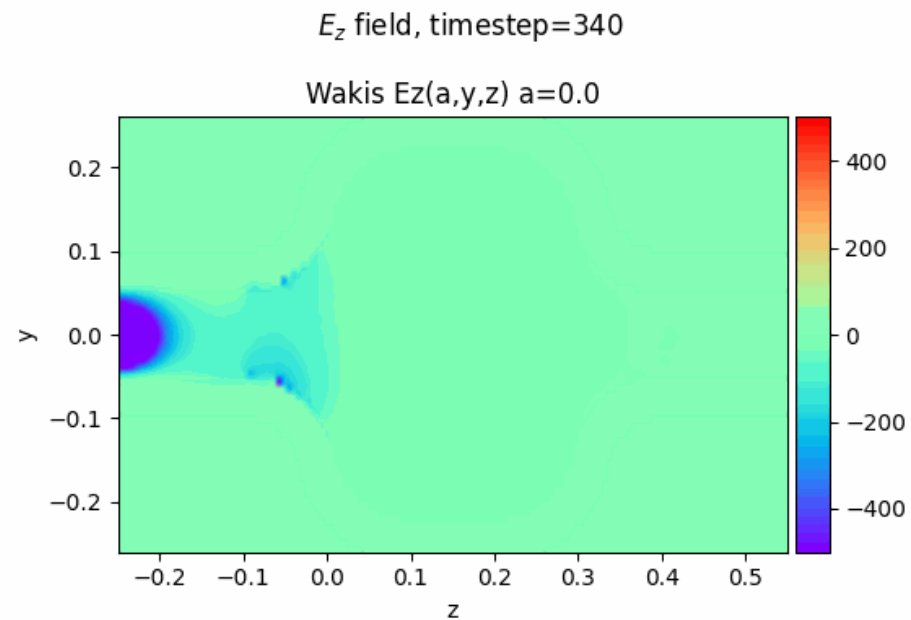


# PBA shells : materials issues: int mask in Solver

Field at plane YZ, x=30

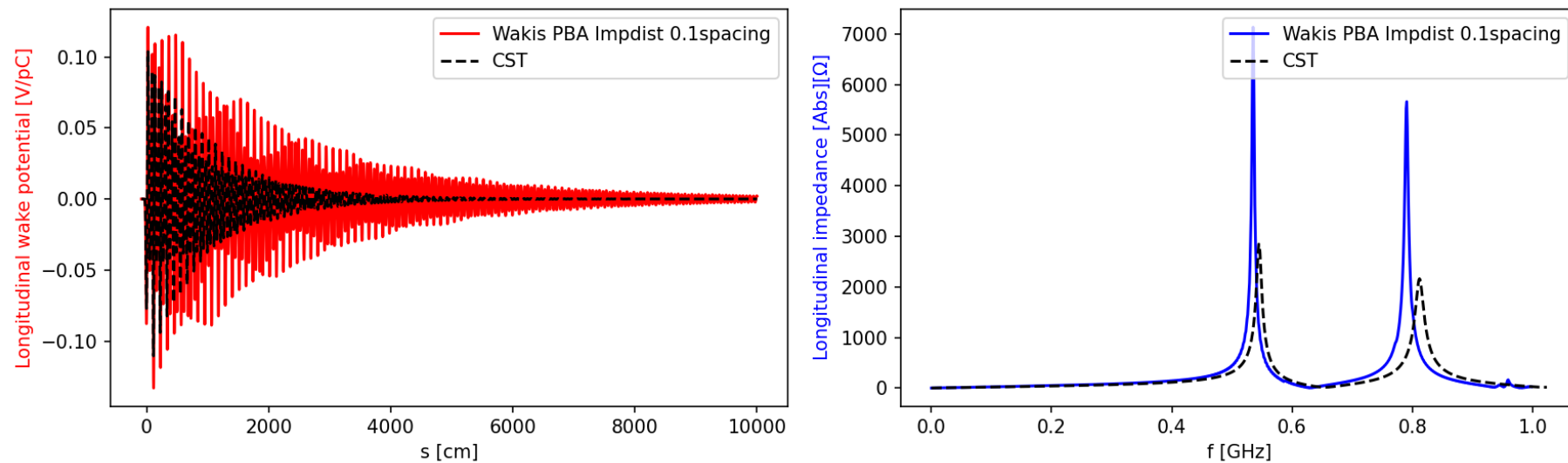


# PBA shells : materials issues: int mask in Solver



# PBA shell 3: simulation results

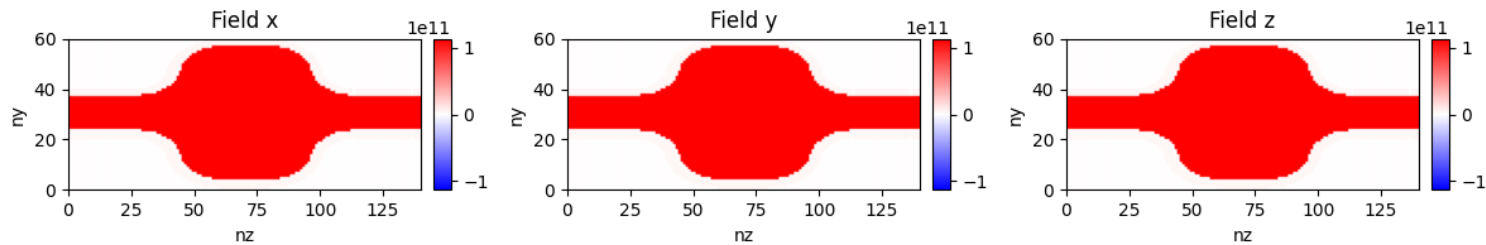
Shell benchmark with CST Wakefield Solver, grid res 60x60x140, cond=30, wl=100m



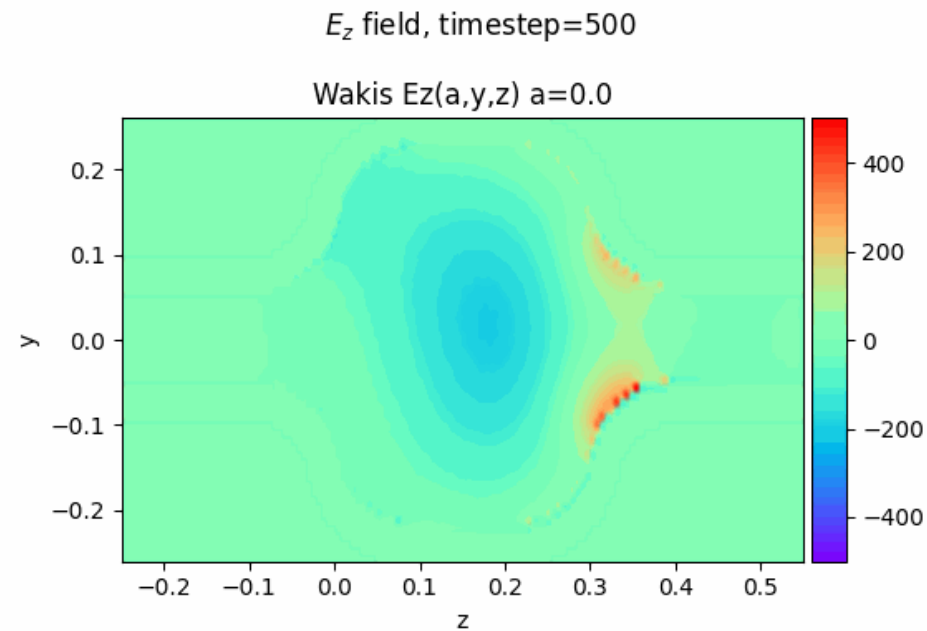


# PBA shells : materials issues: float mask in Solver

Field at plane YZ, x=30



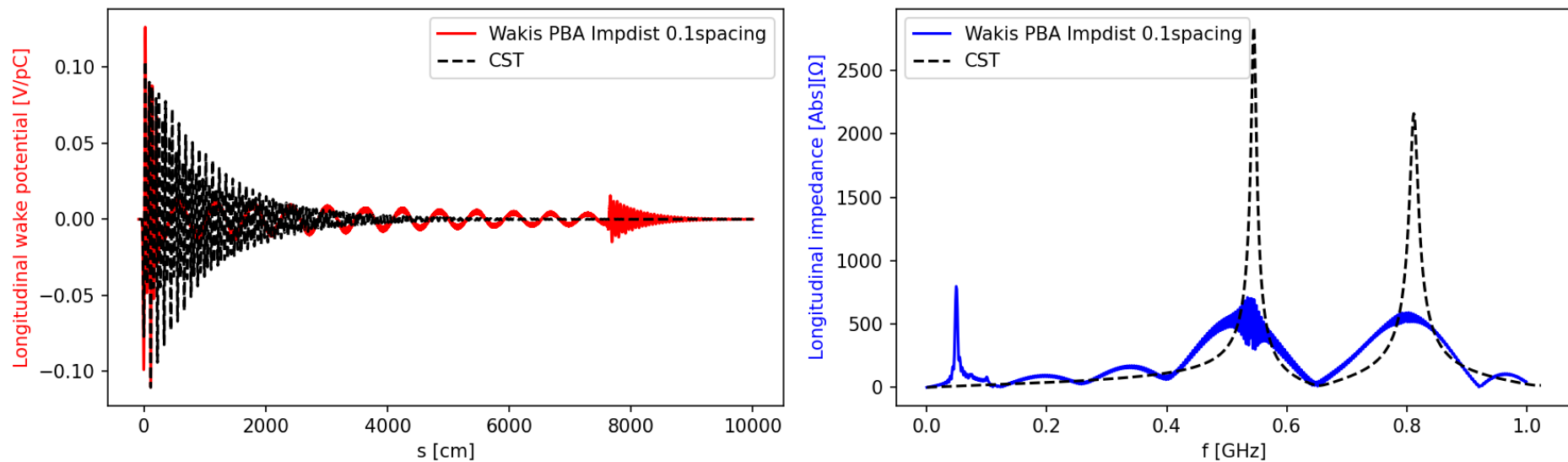
# PBA shells : materials issues: float mask in Solver



# PBA shell 3 (threshold of 0.1): simulation results

## Unstable!! Tried out both bigger and smaller dt

Shell benchmark with CST Wakefield Solver, grid res 60x60x140, cond=30, wl=100m



# Overview slides



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